

Chapter 3 Assignment Due date: May 11, 2026 (Mon)

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3.1

Sol.

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_1 : \mu \neq \mu_0$$

Under H_0 ,

$$\bar{X}_i \sim N\left(\mu_0, \frac{\sigma^2}{m}\right)$$

Hence,

$$Z_i = \frac{\bar{X}_i - \mu_0}{\sigma/\sqrt{m}} \sim N(0, 1)$$

For a two-sided test with significance level α ,

$$P_{H_0}(|Z_i| > z_{1-\frac{\alpha}{2}}) = \alpha$$

Therefore, the rejection region is

$$\begin{aligned} & |Z_i| > z_{1-\frac{\alpha}{2}} \\ \Rightarrow & \left| \frac{\bar{X}_i - \mu_0}{\sigma/\sqrt{m}} \right| > z_{1-\frac{\alpha}{2}} \\ \Rightarrow & \frac{\bar{X}_i - \mu_0}{\sigma/\sqrt{m}} > z_{1-\frac{\alpha}{2}} \quad \text{or} \quad \frac{\bar{X}_i - \mu_0}{\sigma/\sqrt{m}} < -z_{1-\frac{\alpha}{2}} \end{aligned}$$

Since $\sigma/\sqrt{m} > 0$,

$$\begin{aligned} & \bar{X}_i - \mu_0 > z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{m}} \quad \text{or} \quad \bar{X}_i - \mu_0 < -z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{m}} \\ \Rightarrow & \bar{X}_i > \mu_0 + z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{m}} \quad \text{or} \quad \bar{X}_i < \mu_0 - z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{m}} \end{aligned}$$

Let

$$U = \mu_0 + z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{m}}, \quad L = \mu_0 - z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{m}}$$

Therefore,

$$|Z_i| > z_{1-\frac{\alpha}{2}} \iff \text{Reject } H_0 \iff \bar{X}_i < L \quad \text{or} \quad \bar{X}_i > U$$

3.3

Sol.

(i)

Assume

$$\mu_i = \begin{cases} \mu_0, & i < \tau \\ \mu_1, & i \geq \tau \end{cases}$$

where

$$\mu_1 = \mu_0 + \delta, \quad \delta \neq 0$$

From equation (3.4),

$$\hat{\mu}_0 = \bar{\bar{X}} = \frac{1}{n} \sum_{i=1}^n \bar{X}_i$$

Since there are $\tau - 1$ subgroups with mean μ_0 , and $n - \tau + 1$ subgroups with mean μ_1 ,

$$E(\bar{X}_i) = \begin{cases} \mu_0, & i < \tau \\ \mu_1, & i \geq \tau \end{cases}$$

Therefore,

$$\begin{aligned} E(\hat{\mu}_0) &= E\left(\frac{1}{n} \sum_{i=1}^n \bar{X}_i\right) = \frac{1}{n} \sum_{i=1}^n E(\bar{X}_i) = \frac{1}{n} [(\tau - 1)\mu_0 + (n - \tau + 1)\mu_1] \\ &= \frac{1}{n} [(\tau - 1)\mu_0 + (n - \tau + 1)(\mu_0 + \delta)] = \frac{1}{n} [n\mu_0 + (n - \tau + 1)\delta] \\ &= \mu_0 + \frac{n - \tau + 1}{n} \delta \end{aligned}$$

Thus,

$$\text{Bias}(\hat{\mu}_0) = E(\hat{\mu}_0) - \mu_0 = \frac{n - \tau + 1}{n} \delta$$

Since

$$\delta \neq 0 \quad \text{and} \quad 1 < \tau \leq n$$

we have

$$\frac{n - \tau + 1}{n} \delta \neq 0 \Rightarrow E(\hat{\mu}_0) \neq \mu_0$$

Therefore,

$$\hat{\mu}_0 \text{ is not an unbiased estimator of } \mu_0$$

(ii)

For the sampling distribution, assume

$$X_{ij} \sim N(\mu_i, \sigma^2)$$

Then

$$\bar{X}_i \sim N\left(\mu_i, \frac{\sigma^2}{m}\right)$$

Since $\hat{\mu}_0 = \frac{1}{n} \sum_{i=1}^n \bar{X}_i$ and $\bar{X}_1, \dots, \bar{X}_n$ are independent normal random variables,

$$\hat{\mu}_0 \sim N(E(\hat{\mu}_0), \text{Var}(\hat{\mu}_0))$$

Compute

$$Var(\hat{\mu}_0) = Var\left(\frac{1}{n} \sum_{i=1}^n \bar{X}_i\right) = \frac{1}{n^2} \sum_{i=1}^n Var(\bar{X}_i) = \frac{1}{n^2} \sum_{i=1}^n \frac{\sigma^2}{m} = \frac{\sigma^2}{mn}$$

Therefore,

$$\hat{\mu}_0 \sim N\left(\mu_0 + \frac{n - \tau + 1}{n} \delta, \frac{\sigma^2}{mn}\right)$$

3.5

Sol.

$$n = 10, \quad m = 5, \quad \alpha = 0.0027 \Rightarrow z_{1-\frac{\alpha}{2}} \approx 3$$

(i)

$$\begin{aligned} \bar{\bar{X}} &= \frac{1}{10} \sum_{i=1}^{10} \bar{X}_i = \frac{35.53 + 36.01 + 34.74 + 35.30 + 35.80 + 34.41 + 35.35 + 35.19 + 35.05 + 35.66}{10} \\ &= 35.304 \end{aligned}$$

$$\begin{aligned} \bar{R} &= \frac{1}{10} \sum_{i=1}^{10} R_i = \frac{2.41 + 3.24 + 2.24 + 3.16 + 1.06 + 1.80 + 1.60 + 1.42 + 1.77 + 1.42}{10} \\ &= 2.012 \end{aligned}$$

For subgroup size $m = 5$,

$$d_1(5) = 2.326, \quad d_2(5) = 0.864$$

The $\bar{X}$ chart,

$$U_{\bar{X}} = \bar{\bar{X}} + \frac{z_{1-\frac{\alpha}{2}}}{d_1(m)\sqrt{m}} \bar{R} = 35.304 + \frac{3}{2.326\sqrt{5}} \cdot 2.012 \approx 36.465$$

$$C_{\bar{X}} = \bar{\bar{X}} = 35.304$$

$$L_{\bar{X}} = \bar{\bar{X}} - \frac{z_{1-\frac{\alpha}{2}}}{d_1(m)\sqrt{m}} \bar{R} = 35.304 - \frac{3}{2.326\sqrt{5}} \cdot 2.012 \approx 34.143$$

Hence,

$$34.143 < \bar{X}_i < 36.465, \quad i = 1, 2, \dots, 10$$

The R chart,

$$U_R = \left(1 + \frac{z_{1-\frac{\alpha}{2}} d_2(m)}{d_1(m)}\right) \bar{R} = \left(1 + \frac{3 \cdot 0.864}{2.326}\right) \cdot 2.012 \approx 4.254$$

$$C_R = \bar{R} = 2.012$$

$$L_R = \left(1 - \frac{z_{1-\frac{\alpha}{2}} d_2(m)}{d_1(m)}\right) \bar{R} = \left(1 - \frac{3 \cdot 0.864}{2.326}\right) \cdot 2.012 \approx -0.23 \Rightarrow L_R = 0$$

Hence,

$$0 < R_i < 4.254, \quad i = 1, 2, \dots, 10$$

In conclusion, the process seems to be in statistical control.

(ii)

The IC mean is estimated by

$$\hat{\mu}_0 = \bar{\bar{X}} = 35.304$$

The IC standard deviation is estimated by

$$\hat{\sigma} = \frac{\bar{R}}{d_1(m)} = \frac{2.012}{2.326} = 0.865$$

Therefore,

$$\hat{\mu}_0 = 35.304, \quad \hat{\sigma} = 0.865$$

(iii)

When the true process mean is not shifted,

$$\bar{X} \sim N\left(\hat{\mu}_0, \frac{\hat{\sigma}^2}{m}\right)$$

The $\bar{X}$ chart gives a signal if

$$\bar{X} < L_{\bar{X}} \quad \text{or} \quad \bar{X} > U_{\bar{X}}$$

Thus,

$$\begin{aligned} P(\text{signal}) &= P(\bar{X} < L_{\bar{X}}) + P(\bar{X} > U_{\bar{X}}) \\ &= P\left(Z < \frac{L_{\bar{X}} - \hat{\mu}_0}{\hat{\sigma}/\sqrt{m}}\right) + P\left(Z > \frac{U_{\bar{X}} - \hat{\mu}_0}{\hat{\sigma}/\sqrt{m}}\right) \\ &= P(Z < -3) + P(Z > 3) = 2[1 - \Phi(3)] = 0.0027 \end{aligned}$$

(iv)

Under the IC process,

$$ARL_0 = \frac{1}{\alpha} = \frac{1}{0.0027} = 370.37$$

For a mean shift of size 1,

$$\mu_1 = \mu_0 + k\sigma = \mu_0 + 1 \Rightarrow k = \frac{\mu_1 - \mu_0}{\hat{\sigma}} = \frac{1}{0.865} = 1.156$$

Then

$$\bar{X} \sim N\left(\mu_0 + 1, \frac{\hat{\sigma}^2}{m}\right)$$

The probability of no signal after the shift is

$$\begin{aligned} \beta &= P\left(\mu_0 - z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{m}} < \bar{X} < \mu_0 + z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{m}}\right) \\ &= P\left(-k\sqrt{m} - z_{1-\frac{\alpha}{2}} < \frac{\bar{X} - \mu_1}{\sigma/\sqrt{m}} < -k\sqrt{m} + z_{1-\frac{\alpha}{2}}\right) \\ &= \Phi\left(-k\sqrt{m} + z_{1-\frac{\alpha}{2}}\right) - \Phi\left(-k\sqrt{m} - z_{1-\frac{\alpha}{2}}\right) \\ &= \Phi\left(-1.156\sqrt{5} + 3\right) - \Phi\left(-1.156\sqrt{5} - 3\right) \\ &= \Phi(0.415) - \Phi(-5.585) \approx 0.6609 \end{aligned}$$

Therefore,

$$ARL_1 = \frac{1}{1 - \beta} = \frac{1}{1 - 0.6609} = 2.949$$

3.6

Sol.

From Exercise 3.5,

$$\bar{\bar{X}} = 35.304 \quad \text{and} \quad \bar{R} = 2.012$$

$$\bar{s} = \frac{1}{10} \sum_{i=1}^{10} s_i = \frac{0.99 + 1.28 + 0.97 + 1.25 + 0.43 + 0.70 + 0.77 + 0.53 + 0.66 + 0.63}{10} = 0.821$$

For subgroup size $m = 5$,

$$d_1(5) = 2.326, \quad d_2(5) = 0.864, \quad d_3(5) = 0.940$$

(i)

The $\bar{X}$ chart based on s

$$U_{\bar{X},s} = \bar{\bar{X}} + \frac{z_{1-\frac{\alpha}{2}}}{d_3(m)\sqrt{m}} \bar{s} = 35.304 + \frac{3}{0.94\sqrt{5}}(0.821) \approx 36.476$$

$$C_{\bar{X}} = \bar{\bar{X}} = 35.304$$

$$L_{\bar{X},s} = \bar{\bar{X}} - \frac{z_{1-\frac{\alpha}{2}}}{d_3(m)\sqrt{m}} \bar{s} = 35.304 - \frac{3}{0.94\sqrt{5}}(0.821) \approx 34.132$$

Hence,

$$34.132 < \bar{X}_i < 36.476, \quad i = 1, \dots, 10$$

The s chart,

$$U_s = \left(1 + \frac{z_{1-\frac{\alpha}{2}}\sqrt{1-d_3^2(m)}}{d_3(m)}\right) \bar{s} = \left(1 + \frac{3\sqrt{1-0.94^2}}{0.94}\right)(0.821) \approx 1.715$$

$$C_s = \bar{s} = 0.821$$

$$L_s = \left(1 - \frac{z_{1-\frac{\alpha}{2}}\sqrt{1-d_3^2(m)}}{d_3(m)}\right) \bar{s} = \left(1 - \frac{3\sqrt{1-0.94^2}}{0.94}\right)(0.821) \approx -0.076 \Rightarrow L_s = 0 \text{ as } s_i \geq 0$$

Hence,

$$0 < s_i < 1.715, \quad i = 1, \dots, 10$$

In conclusion, the process seems to be in statistical control.

(ii)

The $\bar{X}$ chart based on R ,

$$U_{\bar{X},R} = \bar{\bar{X}} + \frac{z_{1-\frac{\alpha}{2}}}{d_1(m)\sqrt{m}} \bar{R} = 35.304 + \frac{3}{2.326\sqrt{5}}(2.012) \Rightarrow 36.465$$

$$C_{\bar{X},R} = \bar{\bar{X}} = 35.304$$

$$L_{\bar{X},R} = \bar{\bar{X}} - \frac{z_{1-\frac{\alpha}{2}}}{d_1(m)\sqrt{m}} \bar{R} = 35.304 - \frac{3}{2.326\sqrt{5}}(2.012) \Rightarrow 34.144$$

Hence,

$$34.144 < \bar{X}_i < 36.465, \quad i = 1, \dots, 10$$

The R chart,

$$U_R = \left(1 + \frac{z_{1-\frac{\alpha}{2}} d_2(m)}{d_1(m)}\right) \bar{R} = \left(1 + \frac{3 \cdot 0.864}{2.326}\right) (2.012) \Rightarrow 4.254$$

$$C_R = \bar{R} = 2.012$$

$$L_R = \left(1 - \frac{z_{1-\frac{\alpha}{2}} d_2(m)}{d_1(m)}\right) \bar{R} = \left(1 - \frac{3 \cdot 0.864}{2.326}\right) (2.012) \Rightarrow -0.229 \Rightarrow L_R = 0 \text{ as } R_i \geq 0$$

Hence,

$$0 < R_i < 4.254, \quad i = 1, \dots, 10$$

The two $\bar{X}$ charts have limits

$$\bar{X}-s : [34.132, 36.476]$$

$$\bar{X}-R : [34.144, 36.465]$$

These limits are almost identical.

Both the s chart and the R chart show no out-of-control signal.

Both methods suggest that the process is in statistical control

(iii)

Estimators of σ

The range-based estimator is

$$\hat{\sigma}_R = \frac{\bar{R}}{d_1(m)} = \frac{2.012}{2.326} \approx 0.865$$

The sample-standard-deviation-based estimator is

$$\hat{\sigma}_s = \frac{\bar{s}}{d_3(m)} = \frac{0.821}{0.94} \approx 0.873$$

The combined-sample standard deviation is

$$\hat{\sigma}_{\text{combined}} = s_{\text{combined}} = \sqrt{\frac{1}{N-1} \sum_{i=1}^{10} \sum_{j=1}^5 (X_{ij} - \bar{X}_{\text{all}})^2} \approx 0.91$$

The R -based estimator is simple, but it uses only the minimum and maximum values in each subgroup. The s -based estimator uses all observations inside each subgroup and is usually more efficient. The combined-sample estimator uses all observations, it mixes within-subgroup and between-subgroup variation, it may overestimate. For SPC,

$$\hat{\sigma}_R \quad \text{and} \quad \hat{\sigma}_s$$

are more appropriate for estimating within-subgroup variation.

3.7

Sol

The overall probability of Type I error is

$$\tilde{\alpha} = 1 - (1 - \alpha)^n \Rightarrow 1 - \tilde{\alpha} = (1 - \alpha)^n \Rightarrow (1 - \tilde{\alpha})^{1/n} = 1 - \alpha \Rightarrow \alpha = 1 - (1 - \tilde{\alpha})^{1/n}$$

(i)

When

$$\tilde{\alpha} = 0.001 \text{ and } n = 10$$

$$\Rightarrow \alpha = 1 - (1 - 0.001)^{1/10} = 1 - (0.999)^{1/10} \approx 0.0001$$

$$\Rightarrow z_{1-\frac{\alpha}{2}} \approx 3.890$$

When

$$\tilde{\alpha} = 0.001 \text{ and } n = 100$$

$$\Rightarrow \alpha = 1 - (1 - 0.001)^{1/100} = 1 - (0.999)^{1/100} \approx 0.00001$$

$$\Rightarrow z_{1-\frac{\alpha}{2}} \approx 4.417$$

When n increases from 10 to 100, α must become smaller. The $\bar{X}$ chart based on R has limits $U_{\bar{X},R} = \bar{\bar{X}} + \frac{z_{1-\frac{\alpha}{2}}}{d_1(m)\sqrt{m}}\bar{R}$ and $L_{\bar{X},R} = \bar{\bar{X}} - \frac{z_{1-\frac{\alpha}{2}}}{d_1(m)\sqrt{m}}\bar{R}$. Since $z_{1-\frac{\alpha}{2}}$ is larger when $n = 100$, the control limits become wider.

(ii)

When

$$\tilde{\alpha} = 0.0001 \text{ and } n = 10$$

$$\Rightarrow \alpha = 1 - (1 - 0.0001)^{1/10} = 1 - (0.9999)^{1/10} \approx 0.00001$$

$$\Rightarrow z_{1-\frac{\alpha}{2}} \approx 4.417$$

When

$$\tilde{\alpha} = 0.0001 \text{ and } n = 100$$

$$\Rightarrow \alpha = 1 - (1 - 0.0001)^{1/100} = 1 - (0.9999)^{1/100} \approx 0.000001$$

$$\Rightarrow z_{1-\frac{\alpha}{2}} \approx 4.892$$

Compared with part i, $\tilde{\alpha}$ is reduced from 0.001 to 0.0001, so the corresponding α also becomes smaller. The corresponding quantile $z_{1-\frac{\alpha}{2}}$ becomes larger, so the $\bar{X}$ chart limits become wider. Therefore,

$$\tilde{\alpha} \downarrow \Rightarrow \alpha \downarrow \Rightarrow z_{1-\frac{\alpha}{2}} \uparrow \Rightarrow \text{wider control limits}$$

3.8

Sol

(i)

Phase I data:

$$\bar{X}_1, \dots, \bar{X}_{10} = 80.22, 78.31, 81.40, 78.53, 81.32, 80.54, 77.33, 79.24, 81.44, 77.76$$

$$s_1, \dots, s_{10} = 3.99, 5.35, 4.79, 4.68, 3.79, 5.78, 3.52, 4.51, 5.16, 4.81$$

$$\bar{\bar{X}} = \frac{1}{10} \sum_{i=1}^{10} \bar{X}_i = 79.609 \quad \text{and} \quad \bar{s} = \frac{1}{10} \sum_{i=1}^{10} s_i = 4.638$$

For subgroup size $m = 5$,

$$d_3(5) = 0.940, \quad z_{1-\frac{\alpha}{2}} = 3$$

The $\bar{X}$ chart based on s

$$U_{\bar{X}} = \bar{\bar{X}} + \frac{z_{1-\frac{\alpha}{2}}}{d_3(m)\sqrt{m}} \bar{s} = 79.609 + \frac{3}{0.94\sqrt{5}}(4.638) \approx 86.229$$

$$C_{\bar{X}} = \bar{\bar{X}} = 79.609$$

$$L_{\bar{X}} = \bar{\bar{X}} - \frac{z_{1-\frac{\alpha}{2}}}{d_3(m)\sqrt{m}} \bar{s} = 79.609 - \frac{3}{0.94\sqrt{5}}(4.638) \approx 72.989$$

Hence,

$$72.989 < \bar{X}_i < 86.229, \quad i = 1, \dots, 10$$

The s chart

$$U_s = \left(1 + \frac{z_{1-\frac{\alpha}{2}} \sqrt{1 - d_3(m)^2}}{d_3(m)}\right) \bar{s} = \left(1 + \frac{3\sqrt{1 - 0.94^2}}{0.94}\right) (4.638) \approx 9.688$$

$$C_s = \bar{s} = 4.638$$

$$L_s = \max \left\{ 0, \left(1 - \frac{z_{1-\frac{\alpha}{2}} \sqrt{1 - d_3(m)^2}}{d_3(m)}\right) \bar{s} \right\} = 0$$

Hence,

$$0 \leq s_i < 9.688, \quad i = 1, \dots, 10$$

Therefore, the first 10 samples show no out-of-control signal. The process seems to be in statistical control in Phase I.

Then

$$\begin{aligned} \bar{X}_{11} &= 79.48, & \bar{X}_{12} &= 76.74, & \bar{X}_{13} &= 81.12, & \bar{X}_{14} &= 86.79 \\ s_{11} &= 4.56, & s_{12} &= 7.35, & s_{13} &= 3.91, & s_{14} &= 6.31 \end{aligned}$$

For the $\bar{X}$ chart,

$$\bar{X}_{11}, \bar{X}_{12}, \bar{X}_{13} \in [72.989, 86.229] \quad \text{and} \quad \bar{X}_{14} \notin [72.989, 86.229]$$

Thus, $\bar{X}_{14}$ gives an out-of-control signal in the $\bar{X}$ chart.

For the s chart,

$$0 \leq s_{11}, s_{12}, s_{13}, s_{14} < 9.688$$

Thus, no signal is found in the s chart.

(ii)

For the first 10 samples,

$$\overline{s^2} = \frac{1}{10} \sum_{i=1}^{10} s_i^2 = 21.965$$

Since

$$\frac{(m-1)s_i^2}{\sigma^2} \sim \chi_{m-1}^2 \Rightarrow E(s_i^2) = \sigma^2 \Rightarrow Var(s_i^2) = \frac{2\sigma^4}{m-1} \Rightarrow \hat{s}_i^2 = \bar{s}^2 \sqrt{\frac{2}{m-1}}$$

So the s^2 chart

$$U_{s^2} = \left(1 + z_{1-\frac{\alpha}{2}} \sqrt{\frac{2}{m-1}}\right) \overline{s^2} = \left(1 + 3\sqrt{\frac{2}{5-1}}\right) (21.965) \approx 68.561$$

$$C_{s^2} = \bar{s^2} = 21.965$$

$$L_{s^2} = \left(1 - z_{1-\frac{\alpha}{2}} \sqrt{\frac{2}{m-1}}\right) \bar{s^2} = \left(1 - 3 \sqrt{\frac{2}{5-1}}\right) (21.965) \approx -24.631 \Rightarrow L_{s^2} = 0 \text{ as } s_i^2 \geq 0$$

Hence,

$$0 \leq s_i^2 < 68.561, \quad i = 1, \dots, 10$$

The process variability seems to be in statistical control at the first 10 time points

For Phase II, $s_{11}^2 = 4.56^2 = 20.794$ and $s_{12}^2 = 7.35^2 = 54.023$ and $s_{13}^2 = 3.91^2 = 15.288$ and $s_{14}^2 = 6.31^2 = 39.816$

All satisfy

$$0 \leq s_i^2 < 68.561, \quad i = 11, 12, 13, 14$$

Therefore, The s^2 chart gives no out-of-control signal in Phase II

(iii)

Taking the square root of the control limits of the s^2 chart gives a new s chart

$$U_s^* = \sqrt{U_{s^2}} = \sqrt{68.561} \approx 8.280$$

$$C_s^* = \sqrt{C_{s^2}} = \sqrt{21.965} \approx 4.687$$

$$L_s^* = \sqrt{L_{s^2}} = 0$$

Thus, the new s chart is $0 \leq s_i < 8.280$. Square-root s^2 chart: $[0, 8.280]$

The original s chart in part i is $0 \leq s_i < 9.688$. Original s chart: $[0, 9.688]$

The square-root version has a smaller upper control limit.

For Phase I,

$$s_i < 8.280, \quad i = 1, \dots, 10$$

both s charts give no out-of-control signal.

For Phase II,

$$s_{11} = 4.56, \quad s_{12} = 7.35, \quad s_{13} = 3.91, \quad s_{14} = 6.31$$

all satisfy

$$s_i < 8.280, \quad i = 11, 12, 13, 14$$

The square-root version also gives no out-of-control signal in Phase II.

3.9

Sol

$$\bar{X} = \frac{1}{20} \sum_{i=1}^{20} X_i = 24.500$$

(i)

For moving window size $\tilde{m} = 2$

$$d_1(2) = 1.128, \quad d_2(2) = 0.853$$

The moving ranges are

$$\begin{aligned} MR_i &= |X_{i+1} - X_i|, \quad i = 1, \dots, 19 \\ MR_i &= 1, 15, 13, 1, 3, 2, 3, 7, 4, 0, 2, 6, 10, 1, 1, 5, 4, 2, 1 \\ \overline{MR} &= \frac{1}{19} \sum_{i=1}^{19} R_i \approx 4.263 \end{aligned}$$

The $\bar{X}$ chart

$$\begin{aligned} U_{\bar{X}} &= \bar{X} + \frac{z_{1-\frac{\alpha}{2}}}{d_1(m)} \overline{MR} = 24.500 + \frac{3}{1.128}(4.263) \approx 35.838 \\ C_{\bar{X}} &= \bar{X} = 24.500 \\ L_{\bar{X}} &= \bar{X} - \frac{z_{1-\frac{\alpha}{2}}}{d_1(m)} \overline{MR} = 24.500 - \frac{3}{1.128}(4.263) \approx 13.162 \end{aligned}$$

Hence

$$13.162 < X_i < 35.838, \quad i = 1, \dots, 20$$

But

$$X_3 = 39 > 35.838$$

Thus the $\bar{X}$ chart gives an out-of-control signal at X_3 .

The R chart

$$\begin{aligned} U_R &= \left(1 + z_{1-\frac{\alpha}{2}} \frac{d_2(2)}{d_1(2)}\right) \overline{MR} = \left(1 + 3 \frac{0.853}{1.128}\right) (4.263) \approx 13.935 \\ C_R &= \overline{MR} = 4.263 \\ L_R &= \left(1 - z_{1-\frac{\alpha}{2}} \frac{d_2(2)}{d_1(2)}\right) \overline{MR} = \left(1 - 3 \frac{0.853}{1.128}\right) (4.263) \approx -5.408 \Rightarrow L_R = 0 \text{ as } R_i \geq 0 \end{aligned}$$

Hence

$$0 \leq R_i < 13.935, \quad i = 1, \dots, 19$$

But

$$R_2 = |X_3 - X_2| = |39 - 24| = 15 > 13.935$$

Thus the R chart gives an out-of-control signal at moving window R_2 .

(ii)

For moving window size $\tilde{m} = 5$

$$d_1(5) = 2.326, \quad d_2(5) = 0.864$$

The moving ranges are

$$\begin{aligned} MR_i &= \max\{X_i, \dots, X_{i+4}\} - \min\{X_i, \dots, X_{i+4}\}, \quad i = 1, \dots, 16 \\ MR_i &= 15, 17, 17, 5, 7, 7, 7, 7, 12, 10, 10, 10, 10, 5, 5, 5 \end{aligned}$$

$$\overline{MR} = \frac{1}{16} \sum_{i=1}^{16} R_i \approx 9.313$$

The $\bar{X}$ chart

$$U_{\bar{X}} = \bar{X} + \frac{z_{1-\frac{\alpha}{2}}}{d_1(m)} \overline{MR} = 24.500 + \frac{3}{2.326}(9.313) \approx 36.511$$

$$C_{\bar{X}} = \bar{X} = 24.500$$

$$L_{\bar{X}} = \bar{X} - \frac{z_{1-\frac{\alpha}{2}}}{d_1(m)} \overline{MR} = 24.500 - \frac{3}{2.326}(9.313) \approx 12.489$$

Hence

$$12.489 < X_i < 36.511, \quad i = 1, \dots, 20$$

But

$$X_3 = 39 > 36.511$$

Thus the $\bar{X}$ chart gives an out-of-control signal at X_3 .

The R chart

$$U_R = \left(1 + z_{1-\frac{\alpha}{2}} \frac{d_2(m)}{d_1(m)}\right) \bar{R} = \left(1 + 3 \frac{0.864}{2.326}\right) (9.313) \approx 19.690$$

$$C_R = \bar{R} = 9.313$$

$$L_R = \left(1 - z_{1-\frac{\alpha}{2}} \frac{d_2(m)}{d_1(m)}\right) \bar{R} = \left(1 - 3 \frac{0.864}{2.326}\right) (9.313) \approx -1.065 \Rightarrow L_R = 0 \text{ as } R_i \geq 0$$

Hence

$$0 \leq R_i < 19.690, \quad i = 1, \dots, 16$$

All moving ranges satisfy

$$0 \leq R_i < 19.690$$

Thus the R chart gives no out-of-control signal.

For $\tilde{m} = 2$

$\bar{X}$ chart gives a signal at $X_3 = 39$

R chart gives a signal at $R_2 = 15$

For $\tilde{m} = 5$

$\bar{X}$ chart gives a signal at $X_3 = 39$

R chart gives no signal

The major finding is $X_3 = 39$, it is an unusually large observation. The moving window size $m = 2$ is more sensitive to local jumps between adjacent observations. The moving window size $m = 5$ produces a larger average moving range and wider R chart limits.

3.11

Sol

(i)

The sample proportions are

$$p_i = \frac{X_i}{100}, \quad i = 1, \dots, 20$$

The average sample proportion is

$$\bar{p} = \frac{\sum_{i=1}^{20} X_i}{20(100)} = \frac{392}{2000} \approx 0.196$$

The p chart is

$$U_p = \bar{p} + z_{1-\frac{\alpha}{2}} \sqrt{\frac{\bar{p}(1-\bar{p})}{m}} = 0.196 + 3 \sqrt{\frac{0.196(1-0.196)}{100}} \approx 0.315$$

$$C_p = \bar{p} = 0.196$$

$$L_p = \bar{p} - z_{1-\frac{\alpha}{2}} \sqrt{\frac{\bar{p}(1-\bar{p})}{m}} = 0.196 - 3 \sqrt{\frac{0.196(1-0.196)}{100}} \approx 0.077$$

Hence,

$$0.077 < p_i < 0.315 \Rightarrow 7.691 < X_i < 31.509, \quad i = 1, \dots, 20$$

All samples are within the control limits, the process seems to be in statistical control.

(ii)

Assume the IC proportion is $\pi_0 = 0.19$ and the shifted proportion is $\pi_1 = 0.29$.

Using the p chart based on formula (3.19),

$$U_p = \pi_0 + z_{1-\frac{\alpha}{2}} \sqrt{\frac{\pi_0(1-\pi_0)}{m}}$$

To detect the upward shift with probability 0.9, require

$$P_{\pi_1}(\hat{p} > U_p) \geq 0.90$$

We know that

$$\hat{p} \sim N\left(\pi_1, \frac{\pi_1(1-\pi_1)}{m}\right)$$

Thus,

$$\begin{aligned} P_{\pi_1}(\hat{p} > U_p) = 0.90 &\Rightarrow \frac{\pi_1 - U_p}{\sqrt{\frac{\pi_1(1-\pi_1)}{m}}} = z_{0.90} \Rightarrow \frac{\pi_1 - \pi_0 - z_{1-\frac{\alpha}{2}} \sqrt{\frac{\pi_0(1-\pi_0)}{m}}}{\sqrt{\frac{\pi_1(1-\pi_1)}{m}}} = z_{0.90} \\ &\Rightarrow \sqrt{m}(\pi_1 - \pi_0) = z_{1-\frac{\alpha}{2}} \sqrt{\pi_0(1-\pi_0)} + z_{0.90} \sqrt{\pi_1(1-\pi_1)} \\ &\Rightarrow m = \left[\frac{z_{1-\frac{\alpha}{2}} \sqrt{\pi_0(1-\pi_0)} + z_{0.90} \sqrt{\pi_1(1-\pi_1)}}{\pi_1 - \pi_0} \right]^2 \\ &\Rightarrow m = \left[\frac{3 \sqrt{0.19(1-0.19)} + 1.282 \sqrt{0.29(1-0.29)}}{0.29 - 0.19} \right]^2 \approx 309.205 \end{aligned}$$

Hence, the minimum sample size is $m = 310$.

3.13

Sol

$$\sum_{i=1}^{20} X_i = 82 \quad \text{and} \quad \bar{p} = \frac{\sum_{i=1}^{20} X_i}{nm} = \frac{82}{20(50)} = 0.082$$

(i)

$$U_p = \bar{p} + z_{1-\frac{\alpha}{2}} \sqrt{\frac{\bar{p}(1-\bar{p})}{m}} = 0.082 + 3\sqrt{\frac{0.082(1-0.082)}{50}} \approx 0.198$$

$$C_p = \bar{p} = 0.082$$

$$L_p = \bar{p} - z_{1-\frac{\alpha}{2}} \sqrt{\frac{\bar{p}(1-\bar{p})}{m}} = 0.082 - 3\sqrt{\frac{0.082(1-0.082)}{50}} \approx -0.034 \Rightarrow L_p = 0 \text{ as } p_i \geq 0$$

Hence

$$0 \leq p_i < 0.198, \quad i = 1, \dots, 20$$

The process seems to be in statistical control.

(ii)

Because

$$m\bar{p} = 50(0.082) = 4.100 \Rightarrow m\bar{p} < 5$$

The large-sample condition is not satisfied, so use the binomial distribution directly.

Let

$$X \sim \text{Binomial}(50, 0.082)$$

The exact lower limit is

$$L_p^* = \frac{\max\{a : P(X \leq a) \leq \frac{\alpha}{2}\}}{m} = \frac{\max\{a : P(X \leq a) \leq 0.00135\}}{50}$$

Since

$$P(X \leq 0) \approx 0.014 > 0.00135$$

Then

$$L_p^* = 0$$

The exact upper limit is

$$U_p^* = \frac{\min\{a : P(X \geq a) \leq \frac{\alpha}{2}\}}{m} = \frac{\min\{a : P(X \geq a) \leq 0.00135\}}{50}$$

Since

$$P(X \geq 11) \approx 0.002 > 0.00135 \quad \text{and} \quad P(X \geq 12) \approx 0.001 < 0.00135$$

Then

$$U_p^* = \frac{12}{50} = 0.24$$

The exact p chart is

$$L_p^* = 0, \quad C_p = 0.082, \quad U_p^* = 0.24$$

Since

$$0 \leq p_i < 0.24, \quad i = 1, \dots, 20$$

there is no out-of-control signal.

Compared with part (i), the exact binomial chart has a wider upper control limit.

(iii)

The actual Type I error probability is

$$\alpha' = P(X < mL_p^*) + P(X > mU_p^*) = P(X < 0) + P(X > 12) = 0 + P(X \geq 13) \approx 0.000147$$

(iv)

Using the control limits in (3.22)

$$\begin{aligned}\tilde{m} &= m + z_{1-\frac{\alpha}{2}}^2 = 50 + 3^2 = 59 \\ \tilde{p} &= \frac{m\bar{p} + \frac{z_{1-\frac{\alpha}{2}}^2}{2}}{\tilde{m}} = \frac{50(0.082) + \frac{9}{2}}{59} \approx 0.146\end{aligned}$$

The control limit is

$$\begin{aligned}U_p &= \tilde{p} + z_{1-\frac{\alpha}{2}} \sqrt{\frac{\tilde{p}(1-\tilde{p})}{\tilde{m}}} = 0.146 + 3\sqrt{\frac{0.146(1-0.146)}{59}} \approx 0.284 \\ C_p &= \tilde{p} = 0.146 \\ L_p &= \tilde{p} - z_{1-\frac{\alpha}{2}} \sqrt{\frac{\tilde{p}(1-\tilde{p})}{\tilde{m}}} = 0.146 - 3\sqrt{\frac{0.146(1-0.146)}{59}} \approx 0.008\end{aligned}$$

Since

$$0.008 \leq p_i < 0.284, \quad i = 1, \dots, 20$$

there is no out-of-control signal.

In conclusion, All three charts show no out-of-control signal. The (i) gives the narrowest upper control limit. The (iv) gives the widest upper control limit.

3.20

Sol

The target value is

$$T = \frac{USL + LSL}{2} = \frac{2300 + 2100}{2} = 2200$$

(i)

The point estimate of C_p is

$$\hat{C}_p = \frac{USL - LSL}{6s} = \frac{2300 - 2100}{6(50)} = \frac{200}{300} \approx 0.667$$

The point estimate of C_{pk} is

$$\hat{C}_{pk} = \min \left\{ \frac{USL - \bar{X}}{3s}, \frac{\bar{X} - LSL}{3s} \right\} = \min \left\{ \frac{2300 - 2250}{3(50)}, \frac{2250 - 2100}{3(50)} \right\} = \min \left\{ \frac{50}{150}, \frac{150}{150} \right\} = 0.333$$

The point estimate of C_{pm} is

$$\hat{C}_{pm} = \frac{USL - LSL}{6\sqrt{s^2 + (\bar{X} - T)^2}} = \frac{200}{6\sqrt{50^2 + (2250 - 2200)^2}} = \frac{200}{6\sqrt{2500 + 2500}} \approx 0.471$$

Between C_p and C_{pk} , C_{pk} is more appropriate here. Because $\bar{X} = 2250 \neq T = 2200$, the process mean is not centered at the target, C_p only measures spread, while C_{pk} considers both spread and process centering.

(ii)

Since

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2 \quad \text{and} \quad \hat{C}_p = \frac{USL - LSL}{6s}$$

We have

$$P\left(\chi_{\alpha/2, n-1}^2 \leq \frac{(n-1)s^2}{\sigma^2} \leq \chi_{1-\alpha/2, n-1}^2\right) = 1-\alpha \iff P\left(\hat{C}_p \sqrt{\frac{\chi_{\alpha/2, n-1}^2}{n-1}} \leq C_p \leq \hat{C}_p \sqrt{\frac{\chi_{1-\alpha/2, n-1}^2}{n-1}}\right) = 1-\alpha$$

So the 95% confidence interval for C_p is

$$\left[\hat{C}_p \sqrt{\frac{\chi_{0.025, n-1}^2}{n-1}}, \quad \hat{C}_p \sqrt{\frac{\chi_{0.975, n-1}^2}{n-1}}\right] = \left[0.667 \sqrt{\frac{31.555}{49}}, \quad 0.667 \sqrt{\frac{70.222}{49}}\right] \approx [0.535, 0.798]$$

(iii)

We know $X \sim N(2250, 50^2)$ and a defective product occurs when $X < 2100$ or $X > 2300$

Thus

$$\begin{aligned} P(\text{defective}) &= P(X < 2100) + P(X > 2300) = \frac{2100 - 2250}{50} + \frac{2300 - 2250}{50} \\ &= P(Z < -3) + P(Z > 1) = 0.001 + 0.159 \\ &= 0.16 \end{aligned}$$